

AHMED KHALED

Computer Engineering Student | Data Science Track

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[GitHub](#) | [LinkedIn](#) | [Kaggle](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Senior-year Computer Engineering student specializing in Data Science at Ain Shams University. Proficient in C++, Python, and full-stack web development (MERN stack), with hands-on experience in machine learning, embedded systems, and database design. Demonstrated ability to deliver end-to-end software solutions through academic and independent projects. Quick learner and collaborative team member with strong analytical and problem-solving skills.

EDUCATION

B.Sc. Computer Engineering – Data Science Track

Ain Shams University, Faculty of Engineering

Oct 2022 – Present

GPA: 3.6 / 4.0

National Secondary Certificate

Nefertari International School

Sep 2007 – Jul 2022

- Graduated with 93% — Ranked 2nd in class
- Ranked 4th in El-Salam District (2022)

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, Java, C#, Verilog
- **Web Development:** React.js, Node.js, Express.js, MERN Stack, HTML/CSS
- **Databases:** SQL, MySQL, MongoDB, Database Design
- **Machine Learning:** Supervised Learning, Model Training, App Rating Prediction
- **Embedded Systems:** MIPS Architecture, Assembly, Embedded C/C++
- **Tools & Platforms:** Git, GitHub, VS Code, MS Office (Word, Excel, PowerPoint)

PROJECTS

Kora Live

*React, Node.js, PostgreSQL, Redis,
Socket.IO*

- Built a full-stack live football social platform with real-time match reactions and emoji broadcasts via WebSockets, user follow system, and in-app/email notifications
- Implemented a score prediction engine with auto-scoring background jobs (BullMQ) and a global points leaderboard; integrated API-Football for live match data, Cloudinary for avatars, and Resend for email delivery

Syncro – Collaborative Code Editor

*React, Node.js, Yjs (CRDT), Socket.IO,
PostgreSQL*

- Engineered a real-time collaborative code editor using Yjs CRDTs for conflict-free multi-user editing with live cursor tracking, supporting JavaScript, Python, C++, and HTML via CodeMirror
- Implemented room management with public/private rooms, time-limited invite links, join request approvals, file explorer, snapshot history, and ZIP project export; deployed at syncro-ten.vercel.app

Traffic Sign Recognition

Python, OpenCV, scikit-learn, Streamlit

- Developed a classical computer vision pipeline for traffic sign classification on the GTSRB dataset (43 classes): HSV color segmentation, HOG feature extraction, and SVM classifier achieving 90.2% test accuracy

- Built an interactive Streamlit demo app that visualizes each pipeline step (preprocessing, segmentation, feature extraction) and handles real-world images with background clutter

Library Management System

MERN Stack

- Full-stack library platform with admin and student portals, supporting book issuance, reservations, event viewing, and job applications
- Built with React.js, Node.js, Express.js, and MongoDB; team project with 6 contributors

Banking GUI Application

Java, JavaFX, PostgreSQL

- Desktop banking application with full account management, transaction processing (deposits, withdrawals, transfers), and loan lifecycle management
- Implemented admin dashboard with audit reports; wrote unit and integration tests with JUnit 5 and tracked coverage via JaCoCo

Python Compiler

C++

- Built a compiler for a Python subset implementing lexical analysis, parsing, and code generation in C++

Multiplayer Game – Sync Clash

Python

- Developed a networked multiplayer game with real-time synchronization using Python socket programming

University Management System

React.js, Express.js, MySQL

- Full-stack web application for managing students, courses, and faculty; designed relational database schema with MySQL

Reddit Clone

MERN Stack

- Built a full Reddit-like platform with authentication, posts, comments, and voting using MongoDB, Express, React, and Node.js

Door Lock Security System

Embedded C

- Designed and implemented a microcontroller-based door lock system with password authentication

Lint Checker

Verilog

- Created a Verilog linting tool to detect syntax and style issues in hardware description language files

CERTIFICATIONS & TRAINING

- Embedded Systems Diploma – AMIT Learning
- Embedded Systems Training – Siemens EDA Academy
- Big Data Analytics Certificate – National Telecommunications Institute (NTI)
- 2D Game Development – Beginner Training

LANGUAGES

- Arabic – Native
- English – Fluent
- German – Basic

ADDITIONAL INFORMATION

- Military Status: Service Delayed